

GIORGI ARSENADZE

Princeton, NJ / NYC | 646-270-8818 | Email: ga1348@nyu.edu

[Google Scholar](#) | [Github](#) | [LinkedIn](#) | [Personal](#)

PhD candidate in Physics at NYU with research experience in cosmology and biophysics. Experienced in building and testing theoretical models using Python, C++, and numerical simulation tools. Skilled in data analysis and model building across complex scientific systems. Interested in applying computational modeling and data-driven quantitative analysis to study biological systems with a focus on developing tools and simulations that generate experimentally testable scientific insights.

EDUCATION

[New York University](#) *Sep 2019 - Present*

Doctor of Philosophy in Physics

GPA: 3.9/4.0

[New York University](#) *May 2022*

Master of Philosophy in Physics

[Free University of Tbilisi](#) *May 2019*

Bachelors in Physics

EXPERIENCE

Research Assistant in Particle Physics and Phenomenology

in the Group of Prof. Joshua Ruderman

May 2021 – Present

New York University

Developed and tested cosmological dark matter models (“Cannibal Dark Matter” and “Dark Photos”) using Python, C++, and the [CLASS](#) evolution solver. Integrated observational data from cosmological surveys to constrain Dark Matter parameter space, improving existing limits by factors of magnitude. Improved calculation efficiency 10 times through optimized code structure and parallel computation.

Research Assistant in Biophysics

in the Lab of Prof. Alexandra Zidovska

May 2020 – Sep 2022

New York University

Created a data-driven theoretical model to describe the dynamics of nucleoli-droplet-like organelles within human cell nuclei. Built automated image-analysis workflows to process nucleolar data and by combining experimental microscopy data with simulations in Python and MATLAB, reproduced observed experimental behaviors. Demonstrated accurate prediction of droplet coalescence.

Clinical Assistant Professor

April 2023 – Sep 2024

New York University

Led curriculum design and created course work for General Physics I–II, managed and mentored an 8-member teaching team supporting 200+ undergraduates, applied scientifically proven methods to enhance student learning.

Teaching Assistant

Sep 2019 – Present

New York University

Prepared and delivered coursework for 4 experiential and 7 theoretical physics courses, supported faculty with grading and learning management systems.

Undergraduate Researcher

June 2017 – August 2017

Ludwig Maximilian University of Munich

Undergraduate research with Prof. Georgi Dvali on quantum entanglement.

Undergraduate Researcher

Sep 2016 – August 2018

Free University of Tbilisi

Undergraduate research with Prof. Zaza Osmanov studying particle dynamics in a magnetic field in Wormhole Metrics.

Designed advanced curricula emphasizing problem-solving and algorithmic thinking. Coached students to national and international Olympiad awards in Physics and Mathematics. Taught mathematics fundamentals to younger students, fostering early quantitative skills.

PUBLICATIONS

1. **Arsenadze, G.**, Caputo, A., Gan, X., Liu, H., Ruderman, J. T. (2024). **Shaping Dark Photon Spectral Distortions**. Journal of High Energy Physics **2024**
2. **Arsenadze, G.**, Caragine, C. M., Coakley, T., Eshghi, I., Zidovska, A. (2024). **Anomalous Coarsening of Coalescing Nucleoli in Human Cells**. Biophysical Journal. **2024**
3. **Arsenadze, G.**, Osmanov, Z. (2017). **Particles on the Rotating Channels in the Wormhole Metrics**. International Journal of Modern Physics D. **2017**

PRESENTATIONS

1. “Shaping Dark Photon Spectral Distortions” Presentation at the APS Global Physics Summit in Anaheim, CA, March 2025
2. “Constraining Cannibal Dark Matter” Presentation at the APS April meeting in Sacramento, CA, April 2024
3. “Particles on Rotating Channels in Wormhole Metrics” Presentation at the International Conference of Physics Students in Helsinki, Finland, August 2018

WORKSHOPS AND SUMMERSCHOOLS

1. “Particles vs New Probes (P vs NP)”, NYC, USA, 2025
2. “International School of Subnuclear Physics”, Erice, Italy, June 2018

HONORS & AWARDS

Fellowships: JAGA Fellowship (2023–2024); **MacCracken Fellowship** (2019–2024); **Knowledge Fund Fellowship** (2015–2019).

Teaching Awards: **CAS Outstanding Teaching Award** (NYU, 2024); Outstanding Graduate Student Instructor Award (NYU, 2024).

Research & Academic Awards: **Tsinandali Award Laureat** (Nov 2024).

Scientific Competitions: Silver Medal, International Zhautykov Olympiad (**IZhO**) (2015); Honorable Mention, International Physics Olympiad (**IPHO**) (2014, 2015); 2 Gold and 2 Bronze Medals, Georgian National Physics Olympiad (2012–2015).

TECHNICAL SKILLS

Programming:	Python (Pandas, NumPy, SciPy, Matplotlib), MATLAB, C++, Julia, Mathematica, C, Java,
Tools & Systems	Git, High-Performance Computing (HPC), Shell Scripting
Languages	English (Fluent), Georgian (Native), German (Beginner), Russian (Basic)

LEADERSHIP & SERVICE

International Society of Georgian Scientists (ISGS)

Sep. 2020 – Present

Co-founder

Established a nonprofit connecting Georgian scientists worldwide; organized 8 mentorship programs and 20+ webinars, fostering interdisciplinary collaboration and public engagement.

Journal Clubs

Sep 2022 - Present

Co-organizer

Coordinated two journal clubs to discuss recent research, invited guest speakers, and built a community for 30+ graduate students and faculty to explore and exchange scientific ideas.